

Analysis

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Figure 1: Math tech, autumn tea

0 Real Number

0.1 Rational Number Field

Definition 0.1 (Rational Number). A number r is called a **rational number**, if $r = \frac{p}{q}$, where $p, q \in \mathbb{N}$ and $q \neq 0$.

The set of all rational numbers is denoted by $\mathbb{Q}$, and it is called the **rational number field**.

Notes. 1°. **Proof by contradiction.** – To verify if a number is irrational.

Example. Prove that $\sqrt{2}$ is irrational.

Proof. Suppose $\sqrt{2}$ is rational, then $\sqrt{2} = \frac{p}{q}$, where $p, q \in \mathbb{N}$ and $q \neq 0$. Without loss of generality, we assume that p and q are coprime, i.e., $\gcd(p, q) = 1$. Then we have $2 = \frac{p^2}{q^2}$, which implies $p^2 = 2q^2$. Since p^2 is even, p must be even, i.e., $p = 2k$ for some $k \in \mathbb{N}$. Substituting $p = 2k$ into $p^2 = 2q^2$, we have $4k^2 = 2q^2$, which implies $q^2 = 2k^2$. Since q^2 is even, q must be even, which contradicts the assumption that p and q are coprime. Therefore, $\sqrt{2}$ is irrational.

2°. First step of Group, Ring and Field.

Properties. Basic properties of rational numbers:

Order relation:

I-1. (Uniqueness) $\forall a, b \in \mathbb{Q}$, $a < b$, $a = b$, or $a > b$.

I-2. (Transitivity) If $a < b$ and $b < c$, then $a < c$.

I-3. (Density) $\forall a, b \in \mathbb{Q}$, if $a < b$, $\exists c \in \mathbb{Q}$, s.t. $a < c < b$.

(Arithmetic operations)

Addition (& Subtraction):

II-1. (Commutativity) $\forall a, b \in \mathbb{Q}, a + b = b + a$.

II-2. (Associativity) $\forall a, b, c \in \mathbb{Q}, (a + b) + c = a + (b + c)$.

II-3. (Identity0) $\forall a \in \mathbb{Q}, \exists 0 \in \mathbb{Q}$, s.t. $a + 0 = a$.

II-4. (Inverse) $\forall a \in \mathbb{Q}, \exists (-a) \in \mathbb{Q}$, s.t. $a + (-a) = 0$.

II-5. ($>$, $+$) $\forall a, b, c \in \mathbb{Q}$, if $a > b$, then $a + c > b + c$.

Multiplication (& Division):

III-1. (Commutativity) $\forall a, b \in \mathbb{Q}, a \cdot b = b \cdot a$.

III-2. (Associativity) $\forall a, b, c \in \mathbb{Q}, (a \cdot b) \cdot c = a \cdot (b \cdot c)$.

III-3. (Identity1) $\forall a \in \mathbb{Q}, \exists 1 \in \mathbb{Q}$, s.t. $a \cdot 1 = a$.

III-4. (Inverse) $\forall a \in \mathbb{Q}, a \neq 0, \exists a^{-1} \in \mathbb{Q}, \text{ s.t. } a \cdot a^{-1} = 1.$

III-5. (Distribution) $\forall a, b, c \in \mathbb{Q}, a \cdot (b + c) = a \cdot b + a \cdot c.$

III-6. $(>, \cdot) \forall a, b, c \in \mathbb{Q}, \text{ if } a > b \text{ and } c > 0, \text{ then } a \cdot c > b \cdot c.$

Archimedean axiom:

IV-1. (Archimedean property) $\forall c \in \mathbb{Q} \text{ and } c > 0, \text{ then always } \exists n \in \mathbb{N}, \text{ s.t. } n > c.$

0.2 Irrational Number

Introduction. From R.Dedekind, the of defining irrational numbers is very simple, i.e. a (real) number that is not a rational number is an irrational number.

However, it's not so much easy to implement thie idea using mathematical language. For instance, 1. how should we express the stattement "is not a rational number"? 2. on what base should the irrational numbers be constructed? These problems are the mentors who inspire us to enter the rigorous mathematical definitions.

Definition 0.2 (Dedekind Cut). A **Dedekind cut** is a partition of the rational number field $\mathbb{Q}$ into two non-empty sets A and A' , such that

(1) $A \cup A' = \mathbb{Q}$ and $A \cap A' = \emptyset.$

(2) $\forall a \in A, a' \in A', \text{ there must be } a < a'.$

The set A is called the **lower set** and the set A' is called the **upper set**.

Notes. 1°. The condition (1) tells us that $\forall r \in \mathbb{Q}, \text{ there must be } r \in A \text{ or } r \in A'.$

Properties. Observations of Dedekind cut, there are only 3 types of Dedekind cuts:

1 No maximum element in A , but a minimum element r in $A'.$

2 A maximum element r in A , but no minimum element in $A'.$

3 No maximum element in A , and no minimum element in $A'.$

Notes. 1°. No 4th type of Dedekind cut.

Proposition. There doesn't exists a pair of cut, s.t. A has a maximum element and A' has a minimum element.

Proof. Proof is simple, if we notice that the inverse proposition is contract to the condition (1) in definition.

2°. The 3rd Dedekind cut happends to defining the irrational number (we are looking for). 3°. The aforementioned 3 types of Dedekind cut are just theoretical general cases. And actually in different cases, we can construct different Dedekind cut instances.

Definition 0.3 (Irrational Number). Every 3rd type of Dedekind cut corresponds to an **irrational number** $\alpha.$

0.3 Real Number Field

Definition 0.4 (Real Number). A number a is called a **real number**, if a is either a rational number or an irrational number.